

Project Title

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Abstract

Brief summary of the problem, the modelling approach and the main findings.

1 Introduction

1.1 Background and research aims

Describe the system being studied and why it matters. Summarise the relevant literature (for example, Bonabeau, 2002) and state the research question or modelling objective precisely.

1.2 Contribution

Explain what is new in this project relative to existing approaches and the models covered in the unit, and why that contribution provides additional insight.

2 Model specification

Define the entities, state variables and interactions of the model. The update rule for a generic entity is

$$x_i(t+1) = f(x_i(t), \{x_j(t)\}_{j \in \mathcal{N}_i}; \theta), \quad (1)$$

where $\mathcal{N}_i$ is the neighbourhood of entity i and θ collects the model parameters. Explain each term and justify the form of f , referring to the implementation in `src/` of the main repository.

2.1 Assumptions and initialisation

State the modelling assumptions, boundary cases and initialisation rules, and justify the parameter ranges that are used in the experiments.

3 Experimental design

Describe the experiments used to answer the research question: scenarios, parameter sweeps, random seeds, number of repetitions and outcome measures. Summarise them in a table such as Table 1.

Explain the analysis pipeline that turns raw simulation output into the reported statistics, and point to the notebooks and scripts that implement it.

Table 1: Scenarios, parameter ranges and outcome measures.

Scenario	Parameters	Outcome measures
Baseline	values	measures
Variation A	values	measures

4 Results

Present the findings in the order of the experiments, referring to each figure and table from the text.

5 Discussion and conclusions

Interpret the results rather than repeating them: what mechanism explains the observed behaviour, and how does it answer the research question?

State the conclusions that the evidence supports, then discuss limitations, uncertainty and the scope of the model. Identify concrete improvements or applications.

References

Eric Bonabeau. Agent-based modeling: Methods and techniques for simulating human systems. *Proceedings of the National Academy of Sciences*, 99(suppl 3):7280–7287, 2002. doi: 10.1073/pnas.082080899.

A Supplementary material

Optional supporting figures, tables or links (for example, to the committed notebooks and results in the main project repository).